ASTR 352: Astrophysics III

Welcome!

Instructor: Dr. Katie Lester, klester@mtholyoke.edu, 213 Kendade Hall

Please feel free to call me Katie, but I also understand if you are more comfortable using a formal title such as Professor Katie, Professor Lester, ... During the week, I will try to answer any emails within 24 hours. Over the weekend, I may not get back to you as quickly. If you do not hear back from me in two days M-F, please email me again.

Office Hours: I will hold drop-in help hours each week (time TBD), so stop by any time you have questions or want to chat! You're also welcome to email me with questions or to arrange a different meeting time that fits both our schedules.

Overview

Course Description: This advanced course will cover physical processes in galaxies. Topics include the gaseous interstellar medium, dynamics of stellar systems, star clusters, and the viral theorem, along with galaxy rotation, the presence of dark matter in the universe, and spiral density waves. The course concludes with quasars and active galactic nuclei, synchrotron radiation, accretion disks, and supermassive black holes. *Pre-req: ASTR 228*

Course Website: Moodle will contain lecture slides, homework assignments, announcements, grade book, and all other course materials.

Textbook:
– *Extragalactic Astronomy and Cosmology*, by Peter Schneider, 2nd Ed., ISBN 9783642540837. A PDF copy and an eBook copy is available on the Springer website. (PDF also on moodle).
– [optional] *Introduction to Modern Astrophysics*, by Carroll & Ostlie. 2nd Ed.
– [optional] *Galaxies in the Universe*, by Sparke & Gallagher. 2nd Ed.

Grading

Homework (40%)
There will be homework due every 2 weeks that will include qualitative and quantitative questions. You're encouraged to work on assignments with your classmates, but each of you must write your homework in your own words! Make sure to show all of your work for math problems – information that was given, values you're trying to find, equations you could use, steps you took to solve the problem, and checking that your answer makes sense. Please hand write (or digitally write) your homework and turn in a physical copy at the start of class on the due date. Each homework problem

will be graded on a 10-point scale:

10	Excellent	You made no errors and wrote everything neatly & clearly!
8	Good	You've solved the problem, but made some minor mistakes.
6	Poor	You've made a solid attempt at solving the problem, but there are major errors or you missed the point of the question.
0-5	Incomplete	Your answer is incomplete or missing parts of the question.

Once each homework is graded and returned, you have the opportunity to submit a redo to earn one point on the 10-point scale. These redo's have two required parts: (1) redone homework problem(s) that you got wrong, and (2) a short reflection on what went wrong the first time and how you could avoid this mistake in the future. These reflections will help you manage your time better and avoid making the same errors on future homework in addition to helping you think through which concepts you need to review in more detail. These redos+reflections are due anytime before the end of finals, and please turn in the original homework with your redos.

Coding labs & reports (30%)
We will also be doing python coding labs in class, in order to practice coding and applying the concepts you learn in lecture. You will also write up a 1–2 page lab report explaining the methods and results of each lab, due a few days afterwards. This is to practice technical writing and communication skills.

Midterm Project (15%)
Your first project will involve reading & summarizing a recent scientific paper. The midterm projects consists of several parts: worksheet, article draft, peer review, revised article, and presentation. More details about the projects are available on Moodle.

Final Project (15%)
Your second project will be to write a review paper on a topic of your choice (having to do with galaxies). Your paper should be at a higher technical level aimed at other experts in extragalactic astronomy. You'll also give a short presentation in class, in order to practice science communication skills.

Late policy
All assignments turned in after the deadline (without an extension) will be deducted 10% per day with a maximum deduction of 50%. Every assignment has a due date and I expect you to strive to submit each assignment by this date (accounting for any accommodations you have). This ensures I have an opportunity to give you feedback before the next assignment, and missing one often leads to missing another one. Getting behind is overwhelming and can derail your ability to make progress towards our learning goals. I want you to succeed so if you anticipate a problem meeting a due date, email me to propose an extension and we will come to an agreement together. I am always willing to be flexible if you contact me in advance of a due date!

Letter grades

A $\geq$ 94	B+ = 87-89	C+ = 77-79	D+ = 67-69	F $\leq$ 59
A − = 90-93	B = 83-86	C = 73-76	D = 63-66	
	B − = 80-82	C − = 70-72	D − = 60-63	

Learning Goals:

Each assignment you complete in this course will contribute to your growth towards meeting specific learning goals:

1. Demonstrate proficiency in fundamental concepts in each of the major areas of astronomy: cosmology, planetary science, galaxies, stellar structure, and the universe. Show a working knowledge of a broad array of physical phenomena that are based upon fundamental concepts.
2. Demonstrate skills for quantitative analyses, including the ability to form a hypothesis, graphically represent and interpret data, estimate error and understand sampling bias.
3. Gain familiarity with instrumentation, computational methods and software resources utilized by professional astronomers.
4. Demonstrate use of critical thinking skills in well-organized, logical and scientifically sound oral and written scientific reports.
5. Understand the variety of career paths and opportunities that are open to students who have majored in astronomy.

What you can expect from me:

- I will provide you with a clear, organized course that is designed to ensure you meet our learning goals in a meaningful manner.
- I will provide a variety of activities & assignments to ensure your learning needs are met.
- I will provide a supportive and safe environment for you to share and discuss ideas.
- I will treat you with dignity and respect and be flexible to support your individual needs.
- I will reach out to you when I sense that you need support.

What I will expect from you:

- You will strive to be an active participant in this course by coming to class, taking notes, and participating in peer discussion or class activities.
- You will focus on understanding the concepts and performing the skills of this course, aiming for your own personal best.
- You will strive to meet due dates, but will contact me if you have a concern about completing an upcoming assignment.
- You will uphold academic integrity by submitting only work that you understand and completed for yourself.

- You will be thoughtful in your interactions with peers, while taking extra care to respect diverse perspectives. You will support your classmates as you share this learning space and treat them with dignity and respect.
- You will give yourself grace. You will challenge yourself to keep an open mind and recognize that mistakes are a vital part of the learning process.

Class Guidelines

Accommodations
Please let me know if you need any accommodations from the Disability Services office (Mary Lyon Hall 3rd Floor or by email). Send me your accommodations letter, and if needed we can meet to discuss how to apply your accommodations to this class. (For more information on who might be eligible for accommodations and the application process please see the Disability Services website.)

Academic Integrity
I encourage you to work together on homework, since science is often very collaborative. You can help each other with questions and try to figure them out together, but each person must write up their work individually. You are expected to follow MHC's academic integrity policy, and any work that does not will be given a zero and reported to the Academic Honor Board. Each student is required to write up and submit their own work. Using artificial intelligence on assignments is prohibited; students should not have another person/entity do the writing of any portion of an assignment for them, which includes AI tools like ChatGPT.

Mount Holyoke College is a community of students, faculty, staff, and administrators committed to free inquiry and the pursuit of knowledge in the tradition of the liberal arts. The decision to join this academic community requires acceptance of special rights and responsibilities that are essential for its effective functioning and the realization of its mission. All members of the community share the responsibility to uphold the highest standards of academic integrity. I expect all your work to abide by the MHC Honor Code: "I will honor myself, my fellow students, and Mount Holyoke College by acting responsibly, honestly, and respectfully in both my words and deeds." Any work that does not will be reported to the Academic Honor Board.

Tips for success
To help reduce that stress and improve your own likelihood of getting the best grades possible, allow yourself the time and space you need to do your best work. Do not procrastinate and if you get stuck on an assignment, please reach out to me or your classmates. I welcome your questions and I am happy to help you think through your ideas so you can successfully complete an assignment. Sometimes just a quick conversation or email exchange is all you need resolve a problem. Struggle is a natural part of learning, but if you're feeling frustrated that means it is time to reach out for some assistance.

ASTR 352 Schedule - Spring 2026				
Week	Date		Topic	Assignment
1	Wed	Jan 28	Class intro & review	
2	Mon	Feb 2	Coordinates & distances	
	Wed	Feb 4	Milky Way structure	
3	Mon	Feb 9	MW kinematics	HW 1 due
	Wed	Feb 11	Star formation	Worksheet due Fri
4	Mon	Feb 16	Lab 1 - cluster membership	
	Wed	Feb 18	Galaxy classification	Lab 1 report due Fri
5	Mon	Feb 23	Brightness profiles	HW 2 due
	Wed	Feb 25	Galaxy classification	Drafts due Fri
6	Mon	Mar 02	Lab 2 - Hubble's Law	Lab report 2 due Fri
	Wed	Mar 04	Midterm peer review	
7	Mon	Mar 09	Spiral galaxies	
	Wed	Mar 11	Journal club presentations	Midterm article due Fri
8	Mon	Mar 16	No class - spring break	
	Wed	Mar 18		
9	Mon	Mar 23	Elliptical galaxies	
	Wed	Mar 25	Measuring dark matter	HW 3 due
10	Mon	Mar 30	Lab 3 - Andromeda	Outlines due
	Wed	Apr 01	Scaling relations	Lab 3 report due Fri
11	Mon	Apr 06	AGN	
	Wed	Apr 08	AGN	
12	Mon	Apr 13	AGN	HW 4 due
	Wed	Apr 15	Galaxy evolution	Final drafts due Fri
13	Mon	Apr 20	Lab 4 - Reverberation mapping	
	Wed	Apr 22	Final peer review	Lab 4 report due Fri
14	Mon	Apr 27	High redshift universe	
	Wed	Apr 29	Final project presentations	Presentations due
15	Mon	May 04	Final project presentations	
16	Finals period			Final papers due